PETER A. SAYEGH

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EDUCATION

Columbia University | Columbia Engineering — New York, NY
MS in Electrical Engineering – Nikola Tesla Scholarship, GPA: 3.79/4.0 — Expected Dec 2026
Coursework: Digital Signal Processing, Adv. Analog Integrated Circuits, Power Electronics, MOS Transistors, Analog-Digital Interfaces in VLSI, Embedded AI, Microwave Circuit Design, Motor Drive Systems, High-Frequency Magnetics Design
Activities: Lead Department Representative for Electrical Engineering within the Engineering Student Graduate Council (EGSC)

École Polytechnique — Palaiseau, FR
BS in Mathematics and Physics, GPA: 3.75/4.0 — Jun 2025
Coursework: Convex Optimization, Stochastic Processes, Electrodynamics, Adv. Quantum Physics, Thermodynamics, Statistical Physics, Condensed Matter, Algorithms, Numerical Analysis, Web Programming, Object-Oriented Programming, Machine Learning

EXPERIENCE

Columbia Engineering — New York, NY
Course Assistant — Sep 2026 – Present

- Support Prof. Guido Jajamovich and his students in Digital Signal Processing by holding weekly office hours and review sessions and grading homework, exams, and projects

Illumination Diagnostics — New York, NY
Electrical Engineering Intern — Jun – Aug 2026

- Developed a heart-rate extraction pipeline sampling 12 optical channels at 31 Hz, achieving 1–3 bpm agreement against reference
- Prototyped PPG and temperature acquisition systems using ESP32 microcontrollers with synchronized respiration belt recording
- Assembled a 940 nm IR LED illumination array on perfboard (~14 mA/LED) to provide stable background illumination
- Designed a 2-axis pan/tilt laser targeting system (dual servos, Arduino + Waveshare driver), achieving $<0.1^\circ$ angular resolution for programmable wall-target positioning

ELECTRICAL ENGINEERING PROJECTS

Columbia Engineering — New York, NY
Fully Differential Switched-Capacitor Amplifier — Feb – May 2026

- Designed folded-cascode OTA with switched-capacitor CMFB in 0.18 µm CMOS: 56.6 dB gain, 69° phase margin, 487 µW
- Met 0.09% settling error in a 28 ns window for a 7-bit 16 MHz pipelined ADC, at 0.24 pJ/step

Columbia Engineering — New York, NY
250 MS/s 8-bit Current-Steering DAC — Feb – May 2026

- Designed a 3+5 segmented current-steering DAC in 0.18 µm CMOS: 1.53 $V_{pp,diff}$ into 100 Ω at 250 MS/s, 27.6 mW
- Sized 255 unit cells to 0.67% $\sigma(I)/I$ mismatch, bounding 3σ DNL under 0.65 LSB; SFDR $\geq$ 60 dB across corners

Columbia Engineering — New York, NY
SignSpeak: ASL Recognition on Microcontroller — Feb – May 2026

- Deployed a 16K-parameter depthwise-separable CNN on ESP32-S3 (TFLite Micro, INT8): 97.4% accuracy in 67.4 KB flash
- Built a full on-device inference pipeline from camera capture to OLED display, <110 ms latency with no cloud dependency

RESEARCH

École Polytechnique – Center for Theoretical Physics — Palaiseau, FR
Levitation and Stability in Maglev Systems — Feb 2024 – Jul 2026

- Derived a closed-form dipole-layer force model for a scaled maglev prototype, fitting 24 gap-versus-load measurements with R^2 = 0.96 and predicting the unloaded levitation gap within 0.4 mm
- Extrapolated to the Transrapid TR08, showing that a ~1.37 T magnet remanence supports its 574-passenger nominal capacity
- First-author paper published in the *European Journal of Physics* (2026), doi:10.1088/1361-6404/aeaab7

École Polytechnique – Hydrodynamics Laboratory (LadHyX) — Palaiseau, FR
Bachelor's Thesis: Rogue Waves with Physics-Informed Neural Networks — Jan – Mar 2025

- Built a physics-informed neural network in JAX to solve the nonlinear Schrödinger equation for rogue waves, reaching 4.84×10^{-3} average L^2 error with causal training and residual-based adaptive sampling
- Implemented a split-step Fourier spectral solver with $O(dt^3)$ error scaling as the benchmark, validated against analytical dark-soliton solutions with energy conservation error below 10^{-5}

TECHNICAL SKILLS

CAD Tools: Cadence Spectre, Virtuoso, ADE, PLECS, Simulink, LTSpice, Keysight ADS, Xschem, ngspice, KLayout, Ansys Maxwell
Lab and Test: Oscilloscope, function generator, DMM, soldering and rework, perfboard prototyping, benchtop characterization
Programming Languages: Python, C++, Arduino, R, MATLAB, HTML, CSS, JavaScript, QML
Data Science and Machine Learning: TensorFlow, Keras, JAX/Optax, scikit-learn, NumPy, SciPy, pandas, statsmodels, DSPy
Scientific Computing: QuTiP, Scikit-HEP, PySpice
Languages: French (Native), Arabic (Native), German (Advanced), Spanish (Intermediate)